Dr. Christopher A. Tucker

CI/CD - SRE-DevOps in Linux (Azure, OpenShift, microk8s, minikube)
C, Bazel, and C# dotnet development in novel hardware
Polymath

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Profile

Highly-experienced programmer and old-school Linux engineer with deep knowledge of Azure and OpenShift: Containerized deployment strategies, integrated deployments, infrastructure-as-code, docker, constraints, distroless in large and microk8s *at the edge* at zero downtime. I have been an embedded software engineer and signalling specialist as I became cognizant of needing hardware knowledge and must be responsible for my own implementations at an early point in my career, which supplements my profession. I know the old tricks and have mastered them in my daily work. I know many of the old computing languages. Such a foundation has allowed me to express a plethora of skills and techniques to solve a wide spate of technical problems in computing, data, and deployment. Ask me about my work with AiOps!

Languages: English (C2), Dutch (B1), Russian (A1-2) looking to improve my Russian as a part of my profession.

Motto: "*A good guy to have around when things get tough*."

Skills

C, C#, Ruby, Rust, Go, Lua, Lisp, Forth; SVN, BitBucket, Jira, Atlas; Visual Studio, CrossWorks, CodeWarrior; Linux (Debian, RedHat), OS-9, RTOS; gcc, git, bash, az; GitHub Actions, GitLab, Powershell; Azure DevOps, CI/CD, Bazel, Jenkins; Prometheus, Ansible, Splunk, Grafana, Kibana; AKS, Docker, Swarm, Kubernetes, ACR; ArgoCD, SaltStack; Azure Active Directory, Application Gateway, RBAC; Helm, IstIO, Terraform (control planes); Azure Automation, Azure Data Factory; Ansible Playbooks and Jupyter notebooks; HIL Verilog; RS-485, IEEE 1364, IEEE 1783, RS-488, ISO/IEC 14443.

Skill Summary

A well-experienced and educated (both in terms of academia and professional experience) architect and engineer whose focus has been on designing and implementing ideal systems from the developer to production where stable and efficient products is the aim. I design but insist on implementation as I have discovered what I like in a design aligns well with what customers really need. I started as a programmer and am very cognizant of developer needs and methods of guidance without a top-down approach, which has proven itself very successful over the years. A hardcore engineer and old-school programmer by nature, my technical focus is the imperative without losing definition of the problems to be solved by first-principles and team collaboration.

Experience

Azure AKS Subject Matter Expert

June 2024 – Current
Cloudiction, Baarn, The Netherlands

IaC (Terraform, Bicep, ACR, AKS, VNET, RBAC), Azure Devops, GitLab; OPS for development teams transitioning from monolithic applications and outdated VM farm into Kubernetes in a universal manner, but in this case, Azure. Easy-to-use pipelines for devs to reduce stress on transitioning but a long process helping them change the mindset from monolithic to modular deployments.

Technical Design Specialist
May 2022 – October 2023
Dienst Uitvoering Onderwijs, Groningen, The Netherlands

Technical specialist on the CD-Linux team for OpenShift and virtualization where designed highly-granular customizable application deployment environments such as distroless images using Bazel for customized layers via Gitlab, Saltstack, Artifactory, Harbor, and Nexus. Contributor to the ambition of intelligent automated CD systems using artificial intelligence. Results were good in mapping cost in workloads verses business outputs but a notable savings of the hosting by reduction of container footprints using distroless. Kubernetes tailored implementation leveraging GitLab. A result summary is:

- Developed a novel and efficient developer-to-devops method that reduced costs significantly, accelerated developer work, simplified publishing code to the cluster, and reduced security risks making SecOps *very* happy.
- Redesigned and revamped, along with my team, CD1.0 to CD2.0 – from Jboss to OpenShift.
- Was awarded distinctions and recognition of efforts by the director of the organization.
- Implemented best-practices for automation and management of Kubernetes.
- Ansible, Bazel, C#, docker, Kubernetes, Saltstack, bazel, Prometheus, Kibana, Configuration Management, Network, DevOps, CI/CD, PowerShell, Jenkins

CI/CD - SRE in Azure DevOps / Product delivery specialist
August 2021 – May 2022
ASML, Eindhoven, The Netherlands

Hired due to my expertise in CI/CD, developer integration, and Azure architecture and deployment skillset. Duties consisted of planned automation, infrastructure-as-code, pipelines, environments, and Git repos to contain yaml files, Helm, and Terraform control planes and hosting natively in Azure, managed with Powershell; configured monitoring using Prometheus. Additional contribution was toward data engineering, due to my education: pyspark in Databricks for machine parsing of runtime files, including Azure DataFactory. I learned a niche technique at Reading where data can be precompiled and filtered that is analogous to the 2018 concept of "Attention is all you need" but in a more piecewise-linear execution. A result summary is:

- Improved system operating efficiency,
- Solved inherent problems between machine monitoring and software mitigation techniques.
- Introduced new product streams.
- Educated junior data developers to the most efficient ML strategies.
- Bazel, Kubernetes, RestAPI, git workflow, PowerShell, Ansible, SaltStack, Machine Learning, Data Factory

CI/CD - SRE in Azure DevOps / Technical Lead
November 2020 – July 2021
Worth Internet Systems, Rotterdam, The Netherlands

A problem with the nature of resource deployment into Azure Cloud via its DevOps platform prompted them to hire me to help out with several projects in the space of Dutch Ministries. Served as a sparring partner with an architect and project manager tasked to solve a long-running problem of the best approach to modernizing Dutch governmental institutional systems by bringing applications into the modern era with step-by-step procedures. Participated in "pitch sessions" with potential new customers. A result summary is:

- Solved the conundrum by my past experience with DevOps extensions and cataloging.
- Aided in properly configured Azure Gateway and Network Architecture.
- Enhanced my soft skills of talking about technical solutions to non-technical stakeholders.
- Kubernetes, docker, Prometheus, Bazel, Configuration Management, Saltstack, ITIL, Linux, Ansible

CI/CD - SRE in Azure DevOps / Technical Lead

March 2020 – October 2020
Workstreampeople, Rotterdam, The Netherlands

Company is a workflow and infra-communications company that relies on an agent-like approach to customer service. My first task was to build a prototypical call-flow application that could run in a container called *asterisk*. Due to my experience in Azure, I also contributed to helping them bring their server-based applications to the Cloud. Azure did not fit their model in some cases so we did some pilots that could run in their servers in Kubernetes and did some deployment work in Amazon Web Services. Some of the specific technologies utilized in this position was: docker, Kubernetes, asterisk, kamailio, ari.proxy, RBAC, and NATS. Final task was to help them derive a strategy that they could transition away from Jenkins CI/CD toward Gitlab. A result summary is:

- Built a independent module-based container of asterisk that would be deployed when required to pipeline call logic into their management framework.
- Contributed to workloads in AWS.
- Expanded by knowledge of proxy, RBAC, and NATS.
- RHEL, Linux, docker, Kubernetes, PowerShell, ITIL, make, asterisk, bazel, Lisp, Network storage, integration

CI/CD – SRE as Azure Component Specialist / Cognitive Services Technical Lead

February 2019 – March 2020
Tata Steel, IJmuiden, The Netherlands

I was hired to help this industrial company transition their on-premises server-based workflow build systems to the Azure Cloud. The position began in an ad-hoc manner due to fickleness of the Azure architect but due to a lack of understanding in management across the departments that were participating, what was begin could not be undone. Therefore, my task was to find a method that would support on-boarding workloads into the new Cloud system. The solution was to create a DevOps Extension Marketplace consisting as a catalogue offering integrative resource deployments: DTAP environments, namespaces, data-containers (SQL and noSQL), service bus, and Cognitive Services, for Azure deployments utilized within the collective Tata Steel enterprise. A result summary is:

- Resolved a thorny problem as a result of poor practices but keeping the business operating.
- Introduced a novel method to deploy resources by the individual building their specific workload requirements.
- Expanded my knowledge of resources and their deployment paradigm in Azure.
- C#, Json, Kubernetes, PowerShell, ITIL, Network, Gateway, Performance Tuning, Configuration Management, Network storage, integration

C# Senior Software Engineer / Azure Component Specialist (Migration and Integration)

June 2018 – January 2019
Bloemert, Stophorst, The Netherlands

A short-term consulting gig to apply my skills and past code deliveries to improve existing products and infrastructures. Main duties and responsibilities were to advise customers coming from legacy mainframes and networks into the design philosophy of Azure, DevOps, and Agile for best-practises and best results. Worked with NS on the data structure and API between their mobile ride app and the Azure backend. Designed a holistic medical application in C and Java that displayed the operations of a real-time magnetic-resonant scan as it moved through the body of the person receiving the therapy. Spoke about conundrums in software engineering, code size, and artificial intelligence at a sponsored conference in Amsterdam.

C++, C# Senior Software Engineer (Build & Maintenance) / MS Build Technical Lead

June 2015 – September 2017
Shell, The Hague, The Netherlands

Returning to Holland, I was tasked to help evolve the build and maintenance process and procedures of nine teams contributing code to several product-silos of volumetric oil-gas reservoir mapping software first composed in 1997,

as the suite PetroSigns/JewelSuite. Given the long time invested quite a few parts were written in managed C++, some in pure C, and the higher-level abstractions in C#. Some of the technologies that were leveraged during the period were MSBuild, C++ code compilation (on-the-fly), and C# "glue" code to close the gaps between build and product-delivery pipelines all stored in Team Foundation Server. Two separate stints at Shell. A result summary is:

- Devised a compilation method that was triggered by a build pipeline.
- Improved quality by implementing primitive machine-learning algorithms in a build and maintenance context.
- Helped developers understand what goes on behind pushing the F5 button.
- C#, DevOps, Team Foundation, CI/CD, performance tuning, Network, gateway, integrations maintenance

C Software Engineer / Hardware Designer / Technical Lead
January 2014 – February 2015
achelos, Paderborn, Germany

I met an old colleague at the University in Paderborn who told me that this company needed a person with my expertise for a C-language SECCOS EMV for CPA and VCPS for payment infrastructure project for a major German banking-services client. My experience in compilers and system design aided a Renesas-like implementation with functions, select I/O, memory footprint, and parked modes that included: Executive, learning, transactional, and debugging. Given my interest in power systems I introduced into the technical standard of a Berlin-based organization new mathematical models of power transfer for RFID antenna and terminals. A result summary is:

- Expanded my knowledge into micro-miniature CPU platforms.
- Implemented secure code for the SECCOS protocol, based on MiFare.
- Enjoyed the culture in this part of the world.
- Mentored junior colleagues contributing to knowledge growth.
- C, C++, C#, Java, RestAPI, Jenkins, Assembly, Verilog, Ruby, BitBucket

C++ Software Engineer / Firmware Designer
July 2013 – December 2013
Verimatrix, Munich, Germany

Decided to travel through Germany from the southern route. I had done some work in Ruby and decided to take-on some development of it to implement test harness framework in Ruby and Spring framework for a security services company (RSA, AES, TDES) for set-top boxes. Implemented some cryptographic algorithms in C++. In so doing, I was able to improve the scope and quality of an existing Ruby automated test framework for TV subscription boxes.

- Amstrad, Ruby, C++, TDD, BDD, BitBucket, Security make, ldconfig, mdoc, Lisp

Firmware Engineer / Assembly / C-language compilers / Technical Lead
March 2012 – June 2013
NXP, Graz, Austria

I participated in a jobs fair in Amsterdam where Philips/NXP was looking for engineers to work on the MiFare Smart Card chip system, after respite from Virage Logic that helped me to find my way to Europe. NXP had some contractual work at my previous employment, so I was familiar with their innovations and desires to make fast microcomputers at competitive costs. This project was to be a technology that would later find its way into the general transportation sector for contactless consumption of ticketing between check-in and check-out activities across the NS rail system. Their development facility was in Austria, so I relocated to experience this part of Europe. Acquired a deep knowledge of the Chess compiler. Touched upon work from Microware's OS-9.

- Chess, C#, TDD, BDD, make, ldconfig, mdoc, Assembly, C, C++, Forth, Verilog, BitBucket, secure coding, Lisp

Hardware Designer of Metal Architecture / C Programmer
November 2008 – December 2011
Virage Logic, Eindhoven, The Netherlands

I was able to leverage my correspondence with Acron Computers into implementation of chip architecture designs for a variety of embedded contexts. Leaving the UK for Europe for a new adventure found me working in an environment that heralded to the earliest days of microcomputer designs: Notably the i8085, MOS6502, M6809, MC68HC11, and M68000. I was able to form a working understanding of the subtle but significant differences between CISC and RISC architectures and the benefits of each. Worked with assembly and C compilers to leverage different kinds of memory footprints and board designs for manufacture. Worked lastly with ARMv9.0-A.

- SVN, VBScript, VHDL, Verilog, Ruby, Assembly, Altium, C++, NAND Flash, Lisp, Forth, secure code, C

Education

Ph.D. Cybernetics
University of Reading, UK 2008 - 2013

What began as a correspondence with Acron computers (around 2006), the inventors of the ARM architecture, lead to a focus on systems architecture, emphasis on design of die with power aspect. I decided this was outside a pure engineering context as the nature of how the architecture was rendered and the algorithms destined to control it needed investigation. Discovered a close relationship between sinusoidal wireless power and optimized implementation of schemes that predominated the research. It was such a head-spinning exercise that only now am I able to take the time to understand exactly what that meant and how to best leverage it for new designs and products, especially in aerospace. What has always remained relevant was the purpose of architecture by way of physics that determined what designs were successful and other were not: I was able to *discern the 'why' of the problem.*

B.S. Mathematical Physics
University of Utah, USA 2001 – 2004

University was one of those experiences where I did not know what to focus upon. When I was about 12, I discovered a stack of punchcards at my father's desk for an archive project with an IBM1442 and promptly scattered them into disarray just by inspection. My father took the opportunity to not only show me how to order this now archaic technology, how programs were encoded by punches, but also introduced me to C, specifically the *K&R*, 1978 1st edition. These experiences fostered an early interest in microprocessors where a year later I purchased a used Hewlett-Packard 5036A Microprocessor Lab with savings from my paper route. I followed the accompanying book *Practical Microprocessors* getting through most of the lessons and experiments. However, by time for University I was not interested in leveraging this set of skills. One day on a walk I remembered my father advice (more like an epiphany) that if I could not decide, take a degree in physics and work engineering as it can be applied to a variety of practical technical problems facing the world.